Learning Orbital Rendezvous

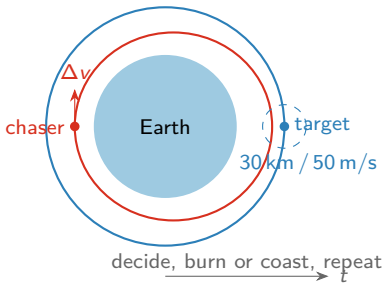
A reinforcement-learned guidance policy, closed through a bearings-only navigation filter

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One policy; seven rendezvous skills in 3-D with J2; 96–99.5% held-out success on the five wide-box skills.
Everything shown was measured on held-out episodes under corrected dynamics.

The problem: get a chaser into a box around a non-maneuvering target



Success $|\mathbf{r}_c - \mathbf{r}_t| < 30\,km$ and $|\mathbf{v}_c - \mathbf{v}_t| < 50\,m/s$

Failure, any of re-entry · escape ($E \geq 0$) · tank empty · clock expires

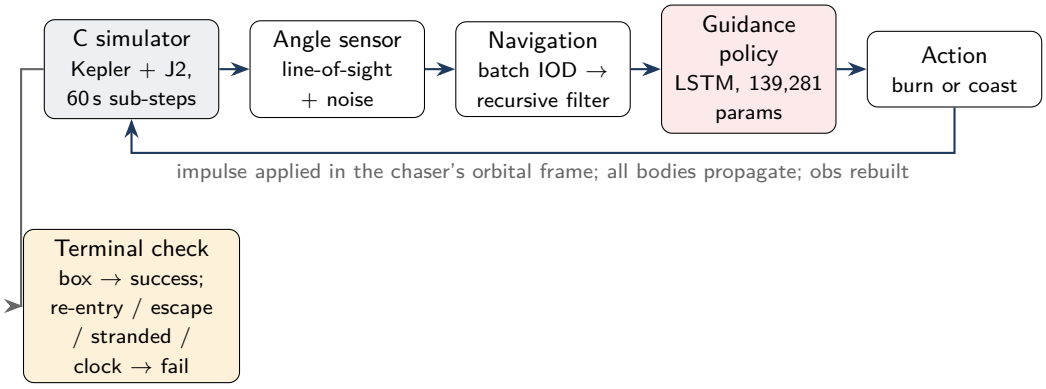
Budgets 478 m/s of Δv (15 % propellant, Isp 300 s, Tsiolkovsky), about 2.2× a Hohmann transfer across the band; a wall clock counted in 60 s sub-steps

Start random LEO orbits, any phase gap up to ±180°, independent eccentricity and apse orientation

Agent sees 38 floats: its own elements, fuel, the *estimated* target, a target-centred relative-motion block, the clock

Agent does one of 31 discrete actions: an impulsive burn, or a coast of 1 min to 24 h

The loop: this is a GNC stack, not a reward-hacking toy



- Guidance never sees truth in the bearings-only configuration; it flies on the filter's estimate.
- The same loop runs in allocation-free C99 on the embedded-style port: bounded execution, action stream identical to PyTorch over 200 episodes.

Simulator: analytic two-body plus J2, written in C

State $(a, e, i, \Omega, \omega, M)$ per body; Cartesian when needed

Propagation $M \mathrel{+}= n\,\Delta t$, then Kepler's equation by Newton–Raphson

$$f(E) = E - e\sin E - M, \qquad E \leftarrow E - \frac{f}{1 - e\cos E}$$

true anomaly from $\tan\frac{\theta}{2} = \sqrt{\frac{1+e}{1-e}}\tan\frac{E}{2}$

Elements ↔ Cartesian perifocal frame, then a 3-1-3 rotation $R_3(\Omega)R_1(i)R_3(\omega)$; the inverse is computed from **h**, **e**, **n** directly

Burns impulsive, decomposed along $\hat{v}$, $\hat{r}$, $\hat{h}$; mass by the rocket equation

J2 secular rates on Ω (node regression), ω (apse rotation), M (clock); the drift the seventh skill exploits

Why analytic

Exact at any step size, so a 24-hour coast costs one Kepler solve, not 1,440 integrator steps. The decision clock and the physics clock are decoupled.

Throughput

Roughly half a million environment steps per second on 1024 parallel envs, one laptop CPU. A 50 M-step training rung is about two hours.

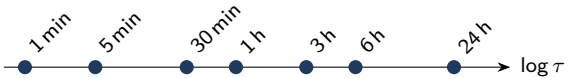
Two-body plus secular J2 only; the state is the mean element set. Earth termination at the surface radius.

Actions: impulses plus variable-length coasts make it a semi-Markov problem

Impulsive burns (one sub-step)



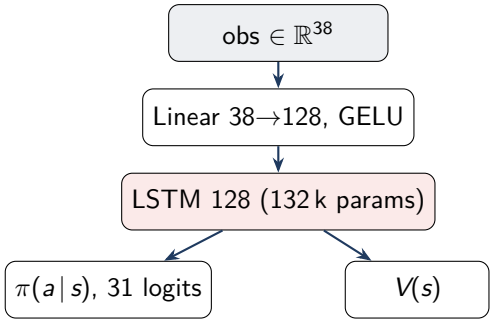
Coasts (no burn, τ sub-steps of 60 s)



physics and sensing still advance every 60 s inside a coast; reward and fuel accrue per sub-step, so warping is never free

- A LEO phasing problem is thousands of minutes long; without coasts the credit-assignment horizon kills learning.
- With coasts the headline policy takes **24 decisions per episode**; the 24 h coast is what lets J2 do a plane change for free.

Policy and training: a small recurrent network trained with PPO



139,281 unique parameters, 557 kB of weights

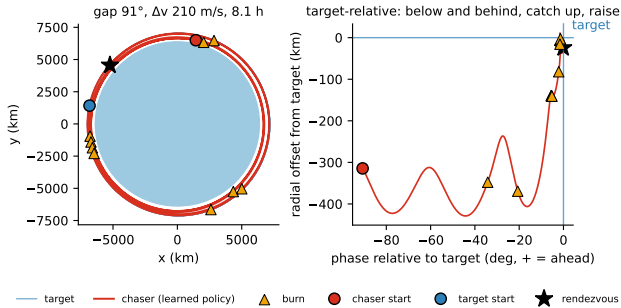
Why an LSTM the estimate's history carries information the instantaneous state does not (is the filter converging? did that burn do what I expected?)

PPO 1024 envs, $\gamma = 0.995$, clipped surrogate ± 0.2, GAE $\lambda = 0.9$, entropy 0.01, Muon optimizer, 50 M steps per curriculum rung (about 2 h each)

Reward $+1$ success, -1 failure, dense potential-difference shaping on the *mean-longitude* gap and element match, clamped to $[-1, 1]$

Curriculum one variable per rung, warm-started: circular at the full $\pm 180^\circ$ gap $\rightarrow$ eccentric $\rightarrow$ wide envelope $\rightarrow$ 3-D $\rightarrow$ estimate in the loop $\rightarrow$ the seven-skill mixture. Fresh training at the hard endpoint flat-lines.

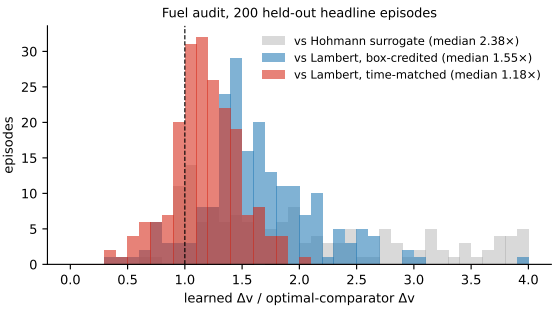
Headline, 2-D: every held-out episode rendezvous, on every training seed



- **200/200** held-out episodes (eval seed 123) for **5/5 training seeds**, including the seed that had flat-lined at 0 % before the dynamics fix.
- Re-run for this deck on a fresh evaluation seed: 200/200.
- Wide envelope (300–8,000 km, $e \leq 0.30$): 4 seeds, 800/800; 3,199/3,200 over the whole four-rung ladder.

Shown: a 91° gap closed in 8.1 h with 210 m/s; low and behind, catch up, then raise.
`models/t3/seed42_L2_headline.pt`.

How good is it? Audit against an optimal-control comparator



Comparator a Lambert solver (Izzo–Lancaster universal variables, multi-rev, both branches) searched over departure wait $\times$ time of flight, with perfect information.

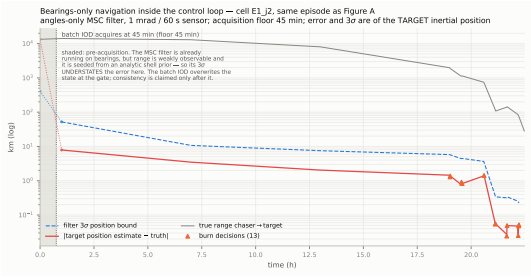
Fair denominator same episode, same time budget: median **1.18**$\times$ the optimum.

Credited to the box 1.55$\times$.

Hohmann surrogate 2.38$\times$, but it ignores phasing and the velocity match, so it was retired as the headline.

200 held-out episodes, `web_data/results/fuel_audit.csv`. Ratios below 1 are possible because arrival is credited inside a box, not at an exact state.

Bearings-only navigation: the policy flies on an angles-only estimate



3-D, cell E1_j2, same episode as the generalist slide. Shaded: blind, angles accumulating. After acquisition the estimate error (red) collapses as range closes; dashed is the filter's own 3σ claim.

Sensor line-of-sight angles only, 1 mrad noise, one per 60 s sub-step; range is never measured

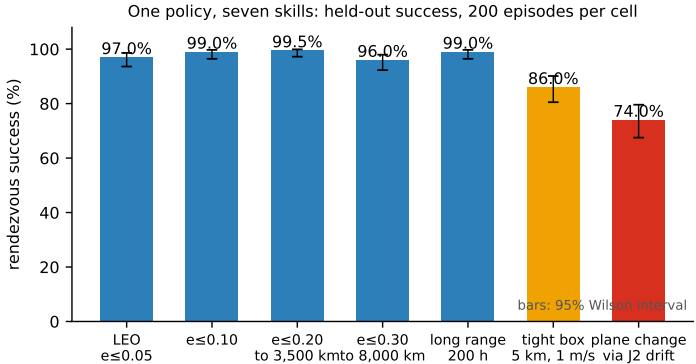
Acquire batch initial-orbit determination: range-hypothesis lattice along the first ray, Levenberg-damped Gauss–Newton, accepted on a χ^2 test and a runner-up margin

Track recursive filter in log-spherical coordinates seeded by the batch solution; calibration checked per channel (a range/bearing EKF with NEES/NIS is in A4)

Result a policy trained on truth and handed the estimate: **51 %**. Retrained with the filter in the loop: **97 %** (2-D tight box, one seed); **46 %→98 %** in 3-D across three seeds.

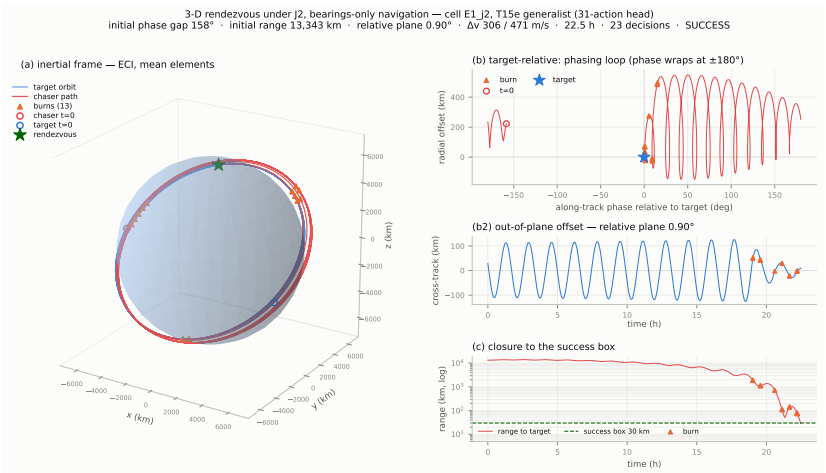
It did not learn to *probe* (measured null); it learned to wait for acquisition and burn nothing while blind (median Δv before acquisition 75→0 m/s).

The generalist: one policy, seven rendezvous skills, 3-D with J2



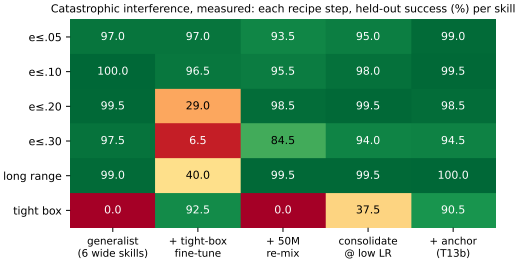
All cells: 3-D, J2, bearings-only navigation, fuel budget sampled per episode, 200 held-out episodes each, one training seed. Tight box reaches 92.5 % at its training fuel floor. Single-skill specialists average 6–42 % off their home cell, which is why one policy was worth building. Cell definitions in A5.

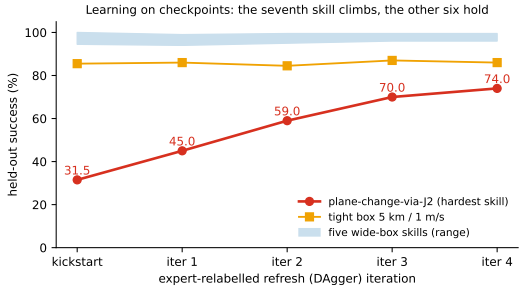
The generalist flying: 3-D, J2, bearings-only, one of the 200 held-out episodes



Cell E1_j2, T15e checkpoint. A 158° gap and a 0.9° plane difference closed in 22.5 h with 306 m/s of 471 and 23 decisions, flying on the filter's estimate. Panel (b) is the phasing loop seen from the target; (b2) the cross-track oscillation the plane burns null at the end; (c) the closure.

How it got there: measuring interference, then fixing it





- Fine-tuning for the tight box destroyed three other skills; re-mixing restored them and lost the tight box. A low-learning-rate consolidation with an **anchor** (replay of the specialist's own trajectories) held all six.
- The seventh skill came from four rounds of **expert-relabelled refresh** (DAgger against a scripted GNC expert), with a stopping rule fixed before the numbers existed. Reproduced at a second seed (33.5 % start, +14 first step).

Verification is the story: what I caught, and what is still open

Caught

- A swapped half-angle factor in the true-to-mean anomaly conversion let every burn teleport phase (median 25° per episode). The old headline fell from 97.5 % to 13 % under correct dynamics; every number here is post-fix.
- A reward leak that made coast-only policies look capable at high altitude (null test: a zero policy reproduced it).
- A red team on the filter: a coordinate-chart swap was a measured null; an unscented filter restored calibration without it.
- A numbered ledger of metric-versus-implementation bugs, each with a before/after number.

Still open

- Generalist and the 2-D 51→97 arm are single-seed; the 2-D headline, wide envelope, 3-D nav and J2 rungs are multi-seed.
- Fuel is 1.18× the Lambert optimum, not 1.0; the action grid is coarse.
- Chaser navigation is assumed perfect; no eclipses, dropouts, or sensor bias.
- Arrival is certified, not station-keeping; two-body plus secular J2 only.

Next

- Relative-state filter with chaser error; measurement gating; multi-seed generalist.
- A matched MPC comparator; a maneuvering target.

Appendix

observation vector · action table · Kepler and Lambert · J2 · filter consistency · noise sweep · the seven cells · embedded port · dynamics bug · red team

A1 Observation vector (38 floats, all roughly in $[-1, 1]$)

slots	value	why
0–1	chaser altitude / scale, e	size and shape
2–3	$\sin\theta, \cos\theta$	position on orbit, no wrap
4–5	v_r/v_{circ}, v_t/v_{circ}	radial and tangential speed
6	fuel fraction	budget remaining
7–8	target altitude, e (estimated)	target size and shape
9–12	$\sin, \cos\ \omega$ for both	apse orientations
13–14	$\sin, \cos\ \Delta\lambda$ (mean longitude gap)	the phase gap, gauge-corrected in 3-D
15	(cap − step)/cap	episode clock
16	$\cos(\omega_s - \omega_t)$	apsidal alignment
17–20	nearest body: distance, bearing, closing rate, keep-out	Earth
21–25	relative-inclination vector in chaser axes; $\Delta\mathbf{e} \cdot \hat{h}_t$; cross-track offset and rate	3-D block
26–28	Δv_{plane}, $\Delta v_{remaining}$, feasibility margin	fuel ledger
29–30	$\cos i_s$, $\cos i_t$	J2 only
33–36	LVLH radial / along-track offset; relative velocity with rotating-frame correction	rectilinear relative state
37	target mean motion	

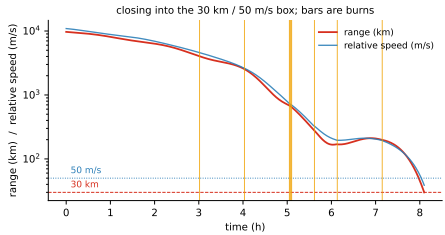
All target quantities are the filter's estimate in bearings-only mode. The policy is not given the covariance (tested: a σ channel added nothing).

A2 Why a lower orbit catches up, and what J2 gives for free

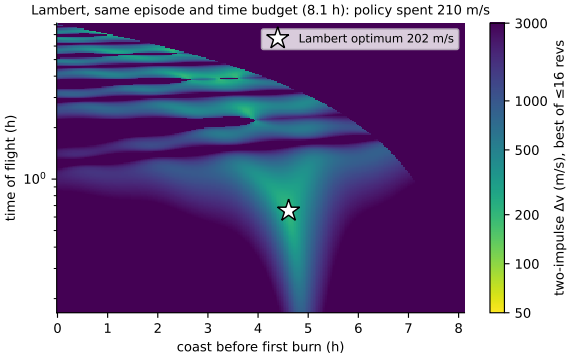
Phasing $T = 2\pi\sqrt{a^3/\mu}$, so a chaser 100 km below a LEO target gains about 4° of phase per orbit. Drop, wait N orbits, raise: the policy's catch-up pattern is the textbook phasing manoeuvre, discovered from reward alone.

Plane change by burning $\Delta v = 2v\sin(\Delta i/2)$; at 7.5 km/s a 3° change costs $\sim$390 m/s, most of the tank.

Plane change by J2 $\dot{\Omega} = -\frac{3}{2}J_2 n(R_E/p)^2\cos i$ depends on a and i, so two orbits at different altitudes drift apart in Ω at a controllable rate. The seventh skill dips low, coasts a day at a time, and lets differential nodal precession close the plane gap (about half the alignment bought this way in the specialist).



A3 Lambert comparator: the search the policy is being graded against



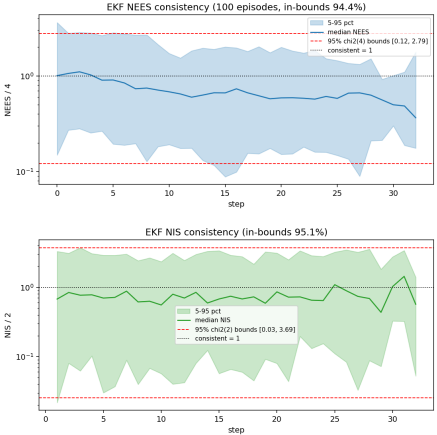
Given $\mathbf{r}_1, \mathbf{r}_2$ and a flight time, find the conic that connects them: Lambert's problem.

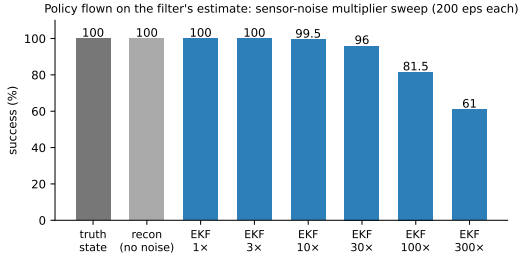
Standard universal-variable form has $A = \sin\Delta\theta\sqrt{r_1 r_2/(1-\cos\Delta\theta)}$, which vanishes at a 180° transfer, exactly the hard phase gap. So the solver uses the Lancaster–Blanchard / Izzo parameterisation in λ and x, with closed-form derivatives of the time-of-flight for a Householder iteration, and multi-revolution branches.

The comparator grids departure wait × time of flight (log-spaced), takes the minimum over revolution counts and branches, then refines around the best cell four times.

This episode: optimum 202 m/s under the policy's

A4 Range/bearing EKF: consistency and sensor-noise robustness (2-D)





NEES $= \tilde{\mathbf{x}}^\top P^{-1} \tilde{\mathbf{x}}$ tests the state covariance; NIS $= \boldsymbol{\nu}^\top S^{-1} \boldsymbol{\nu}$ tests the innovation covariance. Both reported as the fraction of steps inside the 95 % χ^2 band.

2-D range/bearing EKF (σ_ρ 50 m, σ_β 1 mrad) on the headline policy; flat to 10× noise because converged error (tens of metres) is far inside the 30 km box; beyond 30× the limit is filter divergence.

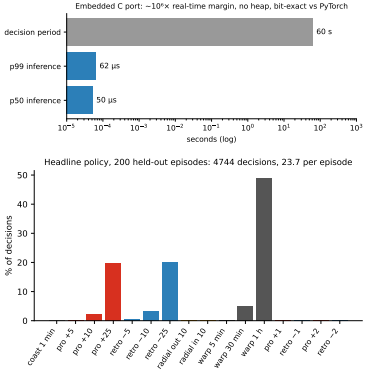
A5 The seven cells of the generalist mixture

cell	a range (km altitude)	e_{max}	box	clock (sub-steps)	weight	success
E0_j2	300–800	0.05	30 km / 50 m/s	3,000	0.15	97.0 %
E1_j2	300–1,500	0.10	30 km / 50 m/s	3,000	0.15	99.0 %
E2_j2	300–3,500	0.20	30 km / 50 m/s	4,500	0.15	99.5 %
E3_j2	300–8,000	0.30	30 km / 50 m/s	6,000	0.15	96.0 %
LONGRANGE	300–3,500	0.12	30 km / 50 m/s	12,000	0.10	99.0 %
TIGHT_5k1	300–800	0.05	5 km / 1 m/s	3,000	0.10	86.0 %[†]
W1_driftwait	300–800	0.05	30 km / 50 m/s	22,000	0.20	74.0 %

[†]92.5 % at the training fuel floor. LONGRANGE is a ∼500 km transfer under a 12,000-sub-step clock. All cells: 3-D, J2 on, bearings-only navigation with real batch acquisition, Discrete(31) head, fuel budget sampled per episode (353–656 m/s) so the policy is fuel-aware. Held-out evaluation seed 123, 200 episodes per cell, greedy action selection, LSTM state zeroed per episode.

Source: `pufferlib/ocean/orbital/t11_cells.py`, `web_data/results/t15e_dagger/fin_*.json`, `T11_GENERALIST.md`.

A6 Embedded C port: the policy as flight software



- No heap: weights are 557 kB of read-only data; recurrent state 1,032 bytes owned by the caller; worst-case stack 3,840 bytes.
- Bounded execution: 138,112 multiply-accumulates per call regardless of input.
- Fail-safe commit: NaN from a bad sensor is rejected, LSTM memory untouched, loop coasts that frame.
- Deterministic, reentrant; ASan and UBSan clean.
- Parity: 100 % action agreement over 7,048 observation vectors; 200/200 closed-loop episodes with an identical action stream to PyTorch.
- p99 62 μs on an M3 Max (a desktop timing, not a WCET) against a 60 s decision period.

A7 The dynamics bug, and why every number here is post-fix

What the true-anomaly → mean-anomaly conversion, called only after a burn, had its $\sqrt{1 \pm e}$ factors swapped. Each burn off an apse silently moved the chaser along-track (up to 700 km at $e = 0.05$); a median 25° of free phasing per episode. Burns at apsides were unaffected, so every unit test passed.

How it was found twice on the same day: by formula inspection while porting the dynamics into the navigation filter, and by the Lambert audit, where 15 % of episodes beat half the time-matched two-impulse optimum. A diagnostic then showed an 86.5° true-anomaly jump for a 25 m/s burn against a 1 km displacement.

Cost the then-canonical checkpoint went from 97.5 % to 13 % under corrected dynamics. All pre-fix checkpoints and numbers were retired.

Recovery recipe (each piece ablated)

- potential shaping on the *mean-longitude* gap with $\gamma_{shape} = 1$
- physical-gap initialisation and a fixed phase observation
- an episode-clock observation; cap reward 0 not −10
- scripted GNC expert as a reference: 99.2 %

Lesson the comparator caught what the success metric could not. Audits against physics are part of the training loop, not a report at the end.

A8 Red team on the navigation filter

- **Hypothesis** the EKF's coordinate chart (Cartesian) was causing the acquisition-stage fragility; a modified-spherical-coordinates chart should fix it.
- **Test** chart swapped in the closed loop, 2-D and 3-D, same seeds, same episodes.
- **Result** a measured null: 2-D $-1/0/0/-4/+1$ per 200 across five arms; 3-D -0.17 pp with a $[-1.2, +0.8]$ interval. Production filters reproduced the prototype's fragility; an unscented filter recovered calibration without a chart change.
- **What actually fixed acquisition** batch initial-orbit determination before handing off to the filter (never-acquire episodes $44/200 \rightarrow 0/200$).
- **Why report it** five claims in my own interview notes were contradicted by this test. The document was corrected.

`reports/msc_REDTEAM.md` §9.

A9 If asked

Why RL at all, when Lambert exists? Lambert needs perfect state and gives an open-loop plan; the policy is closed-loop on an estimate, handles a budget, and the same network covers seven regimes. Lambert is my grader, not my competitor.

Why discrete actions? PPO is stable with them; a continuous Box(2) head collapsed in early trials. The grid costs fuel (1.14–1.18$\times$ the time-matched optimum across two evaluation sets); finer grades are the obvious next step.

Why not MPC? It would be the right production baseline. The project's question was whether a learned policy can close the loop on estimated state across regimes; MPC with the same filter is the comparison I would run next.

Is 200 episodes enough? For a 97 % point the Wilson 95 % interval is about ± 2.5 pp; for 0/200 the two-sided Clopper–Pearson lower bound is 98.2 %. I report intervals, and I do not say 100 % means zero.

Single seed? The generalist, yes. The 2-D headline (5 seeds), wide envelope (4), 3-D nav rungs (3), J2 tight box (3) are multi-seed.

Why does the policy coast 24 h? Because J2 node regression closes an inclination gap for free; burning it would exceed the tank. It learned that.

What would you do differently? Audit against physics from day one; log learning curves on every run; design the observation with the filter in mind from the start.